Shape and Space 3 — The tangent ratio · Angles of elevation and depression · Bearings

Calculator allowed. Give answers correct to 3 significant figures. Draw a labelled right-angled triangle for every question. Marks in brackets. ★ = harder question.

1 In a right-angled triangle, the side adjacent to a $38°$ angle is 14 cm long. Find the length of the side opposite the $38°$ angle. [3]

> $\tan 38° = \dfrac{\text{opp}}{\text{adj}} = \dfrac{x}{14} \Rightarrow x = 14\tan 38° = 10.9$ cm.

2 An aeroplane takes off and climbs in a straight line. When it has travelled 600 m horizontally it is 250 m above the ground. Calculate its angle of elevation θ from the take-off point. [3]

> $\tan\theta = \dfrac{250}{600} \Rightarrow \theta = \tan^{-1}\left(\dfrac{250}{600}\right) = 22.6°$.

3 From the top of a lighthouse, the angle of depression of a small boat is $18°$. The boat is 150 m from the base of the lighthouse. Find the height of the lighthouse. [4]

> The angle of depression from the top equals the angle of elevation from the boat (alternate angles), so in the right-angled triangle: $\tan 18° = \dfrac{h}{150} \Rightarrow h = 150\tan 18° = 48.7$ m.

4 A hot-air balloon rises vertically from a point on the ground. An observer stands 280 m away. When the balloon stops for the first time, its angle of elevation from the observer is $30°$; when it stops at its highest point the angle is $55°$. [7]

a Find the height h_1 of the first stop and the further height h_2 it then rises.

b The balloon takes 1 minute to reach the first stop and a further 6 minutes to reach the top. Calculate the average speed of the whole ascent in m/s.

> **a** $h_1 = 280\tan 30° = 161.7$ m. Total height $= 280\tan 55° = 399.9$ m, so $h_2 = 399.9 - 161.7 = 238$ m (3 s.f.).
>
> **b** Total ascent 399.9 m in 7 min $= 420$ s: $\dfrac{399.9}{420} = 0.952$ m/s.

5 A ship sails 8 km due north from a port P, then 6 km due east to a point Q. [8]

- **a** Draw a diagram of the route.
- **b** How far is Q from P in a straight line?
- **c** Calculate the bearing of Q from P.
- **d** Write down the bearing of P from Q.

> **a** A right-angled triangle: 8 km up (north), then 6 km to the right (east); PQ is the hypotenuse.
> **b** Pythagoras: $\sqrt{8^2 + 6^2} = 10$ km.
> **c** The angle at P between north and PQ: $\tan\theta = \dfrac{6}{8} \Rightarrow \theta = 36.9°$. Bearing of Q from P is $037°$ (three figures, clockwise from north).
> **d** The reverse bearing: $36.9 + 180 = 216.9 \approx 217°$.

6 A vertical mast stands on level ground. From a point 40 m from its base, the angle of elevation of the top is $35°$. From a point 25 m further away, in the same straight line, find the angle of elevation of the top. [4]

> Height of mast: $40\tan 35° = 28.0$ m. From 65 m away: $\tan\theta = \dfrac{28.0}{65} \Rightarrow \theta = 23.3°$ (keep the unrounded height on the calculator).

Total: 29 marks

Check: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$ — label the sides from the angle you're using · angle of depression = angle of elevation from the other end · bearings: from north, clockwise, three figures · reverse bearing: ± 180°.